

Core A' — the nineteen phases, in the order spec §11 puts them

6 merged, 13 to go · 54 of 150 tasks ticked · as of 2026-09-05 · the third line of each chip is what the phase puts on fig 1d

the framework
no syn3A biology at all — D0's order, and every phase merged so far except 5b, which is last in the band and is what unblocks the fan-out

✓ phase 0 · #41
Retire OpenSpec, keep its findings
— nothing in the figure —
merged 2026-09-03 · 6/6 tasks

✓ phase 1 · #42
A module contributes to a state it does not own
the mass and currency arrows inside the ODE grid
merged 2026-09-03 · 7/7 tasks

✓ phase 2 · #43
Several jump modules compose
the CME block being three coexisting boxes at all
merged 2026-09-03 · 7/7 tasks

✓ phase 3 · #44
The 1 s handshake, on a two-module toy. The kill phase
the HOOK device · catalytic and deferred-counter arrows
merged 2026-09-05 · 9/9 tasks

✓ phase 4 · #45
The 60 s rebuild
the REBUILD device · the rate-constant arrows
merged 2026-09-05 · 7/7 tasks

✓ phase 5 · #46
Growth and volume
the GROWTH device · the volume arrow
merged 2026-09-05 · 7/7 tasks

□ phase 5b
The inbound volume channel
the volume arrow's inbound half — geometry into a rate law
11/11 tasks, not merged

the fan-out — ODE block
6, 7 and 8 are mutually independent; 9 waits on 8

□ phase 6
Central glycolysis
box: Central (10 rxns)
7 tasks, none ticked

□ phase 7
Phosphotransferase transport and lactate export
box: Transport (6 rxns)
7 tasks, none ticked

□ phase 8
Nucleotide recycling
box: Nucleotide (5 rxns)
8 tasks, none ticked

□ phase 10
Transcription
box: transcription (17 genes)
8 tasks, none ticked

□ phase 9
Lumped tRNA charging, in the ODE block
box: tRNA charging (the box that moved)
10 tasks, none ticked

□ phase 11
Translation
box: translation (17, plus ptsG translocation)
11 tasks, none ticked

□ phase 12
Transcript decay
box: mRNA decay (17)
8 tasks, none ticked

□ phase 13
Assemble Core A' and assert structural completeness
the whole figure asserted complete, as an object in code
8 tasks, none ticked

□ phase 14
The validation checks, in the scoping note's order
conservation checks over the figure
10 tasks, none ticked

□ phase 15
Observables and synthetic data
— nothing in the figure —
8 tasks, none ticked

□ phase 16
Recovery, coverage, and the reference
— nothing in the figure —
11 tasks, none ticked

□ phase 17
Per-module calibration
— nothing in the figure —
tasks are written once phase 16 has run

the fan-out — CME block
10 before 11 and 12, which both read the transcripts 10 owns

assembly, validation, inference
strictly serial — each check catches what the next would mask

Every merged phase is framework, and that is D0 on purpose — the protocol changes first, then the kill risk on a two-module toy, before a line of syn3A biology. It is also why fig 1d looks untouched: the phases that draw boxes all lie ahead. Phases 15–17 put nothing on fig 1d at all. They are the layer the project exists for, and a state graph cannot show it.